COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 2 : Session – 2 : SLO – 2 | Operational Concepts
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

Section A – Match the Following (Answers)

• 1. PC (Program Counter) → C. Points to the next instruction to be executed
• 2. MAR (Memory Address Register) → D. Holds address for accessing memory
• 3. MDR (Memory Data Register) → E. Holds the fetched data or instruction
• 4. ALU (Arithmetic Logic Unit) → A. Performs arithmetic and logic operations
• 5. IR (Instruction Register) → B. Temporarily holds instruction fetched from memory

Section B – Fill in the Blanks

• At the start of execution, the PC (PROGRAM COUNTER) is set to point to the first instruction.
• The address from the PC is loaded into the MAR, which initiates a read operation.
• The fetched instruction is placed into the IR for decoding.
• The ALU performs the operation, and the result is written back using the MAR and MDR.
• An INTERRUPT is a service request by an I/O device to the processor.

Section C – Activity Task: Program Execution Cycle

• 1. Program load via input device: the program and its data are transferred from an input device into main memory.
• 2. Fetch cycle (PC → MAR → Memory → MDR → IR): PC sends the instruction address to MAR; memory is read; the instruction travels through MDR into IR.
• 3. Decode instruction: the Control Unit examines IR, identifies the operation and the operands needed.
• 4. Fetch operands: operand addresses are sent to MAR and the data is read into MDR/registers.
• 5. ALU execution: the ALU performs the required arithmetic or logic operation on the operands.
• 6. Write back result: the result is sent through MDR to memory or into a destination register.
• 7. PC increment: PC is updated to point to the next instruction and the cycle repeats.

Section D – True or False

• The PC is used to perform arithmetic operations. — FALSE
• The Instruction Register (IR) holds the address of data. — FALSE (it holds the instruction itself)
• Interrupts help the processor handle I/O device requests. — TRUE
• ALU operations are performed before fetching the instruction. — FALSE
• After an interrupt is serviced, the processor resumes the program from where it was interrupted. — TRUE